

Object-Oriented Software Engineering

التعريفات:

الصريح منها على الأقل.

Software Engineering: The process of solving customers' problems by the systematic development and evolution of large, high-quality software systems within cost, time and other constraints.

Or: The application of a systematic, disciplined, quantifiable approach to the development, operation, maintenance of software; that is, the application of engineering to software.

Or: The systematic activities involved in the design, implementation and testing of software to optimize its production and support

Design (as a software activity): Deciding how the requirements should be implemented, using the available technology.

Modeling (as a software activity): Creating representations of the domain or the software.

Object Oriented paradigm: An approach to the solution of problems in which all computations are performed in the context of objects.

Object: A chunk of structured data in a running software system.

Class: A unit of abstraction in an object oriented (OO) program.

Variable: Refers to an object.

Operation: A higher-level procedural abstraction that specifies a type of behavior.

Method: A procedural abstraction used to implement the behavior of a class.

Polymorphism: A unit of information carriage.

Inheritance: The implicit possession by all subclasses of features defined in its superclasses.

framework: is reusable software that implements a generic solution to a generalized problem.

distributed system: a system in which computations are performed by separate programs normally running on separate pieces of hardware that co-operate to perform the task of the system.

Server: A program that provides a service for other programs that connect to it using a communication channel.

Client: A program that accesses a server (or several servers) to obtain services.

Domain Analysis: The process by which a software engineer learns about the domain to better understand the problem.

Requirements document: A collection of requirements.

Requirement: is a statement describing either an aspect of what the proposed system must do, or a constraint on the system's development.

use case: is a typical sequence of actions that a user performs in order to complete a given task.

Scenario: is an instance of a use case.

UML: The Unified Modelling Language is a standard graphical language for modelling object oriented software.

Propagation: A mechanism where an operation in an aggregate is implemented by having the aggregate perform that operation on its parts.

Object Constraint Language (OCL): a specification language designed to formally specify constraints in software modules.